

09 - Bedside Synthesis

Companion reading handout for simultaneous listening.

Course: Medical Prevention of Spontaneous Early Delivery

The course ends where the patient begins.

A resident does not need a floating list of trials. A resident needs a way to stand in front of a patient with a short cervix or a prior spontaneous preterm birth and know what kind of reasoning is required.

Use this sequence.

First, define the phenotype.

Is this a singleton pregnancy? Is the patient asymptomatic? Are the membranes intact? Is there cervical dilation? Is there a prior spontaneous preterm birth? Is this one prior event or two? Were the prior births late preterm or very early? Is this a current short cervix without prior history, or recurrence risk plus current cervical shortening?

Second, define the timing.

At what gestational age was the cervix measured? Before twenty-four weeks, the patient may resemble the strongest short-cervix progesterone evidence. After twenty-four weeks, especially after twenty-eight weeks, late initiation becomes a different evidence question.

Third, define the endpoint.

Are we trying to reduce birth before thirty-seven weeks? Before thirty-four? Before thirty-three? Before twenty-eight? Are we trying to gain days from diagnosis to delivery? Are we trying to improve neonatal outcomes? Do not say "early delivery" unless you can name the endpoint.

Fourth, map the patient to the evidence.

If she is asymptomatic, singleton, no prior spontaneous preterm birth, and the cervix is twenty millimeters or less before twenty-four weeks, the guideline-supported medical treatment is vaginal progesterone. If the cervix is twenty-one to twenty-five millimeters before twenty-four weeks, the decision is more preference-sensitive. If the cervix is first found short after twenty-four weeks, the evidence for starting treatment is weaker. If she has prior spontaneous preterm birth without short cervix, vaginal progesterone is not simply a replacement for 17-OHPC.

Now test the sequence with three patients.

Patient one: asymptomatic singleton pregnancy, no prior spontaneous preterm birth, cervical length nineteen millimeters at twenty-one weeks.

Phenotype: asymptomatic singleton, no prior preterm birth. Timing: before twenty-four weeks. Endpoint: prevention of early spontaneous preterm birth, especially before thirty-four or thirty-three weeks. Evidence match: strong. This patient is close to the clean short-cervix progesterone evidence and the SMFM recommendation zone.

The counseling can be confident but still exact: "The strongest evidence applies to patients like you, with a short cervix found before twenty-four weeks. Vaginal progesterone reduces early preterm birth in this phenotype. The recommendation is strongest at twenty millimeters or less."

Patient two: asymptomatic singleton pregnancy, no prior spontaneous preterm birth, cervical length twenty-two millimeters at twenty-six weeks.

Phenotype: asymptomatic singleton, no prior preterm birth. Timing: after twenty-four weeks. Endpoint: now time gained and prevention of later threshold births become central, because some earlier endpoints are already passing. Evidence match: weaker.

This is not the same patient as the first one. The cervical length is abnormal, but the initiation evidence has changed. Late-initiation studies in no-prior-preterm-birth populations have not shown clear reduction in spontaneous preterm birth or prolongation of pregnancy. A decision to treat, if made, should be described as uncertain or extrapolated rather than proven.

Patient three: singleton pregnancy, prior spontaneous birth at twenty-eight weeks, now a short cervix is found.

Phenotype: recurrence risk plus current cervical shortening. Timing must be defined. Endpoint must be defined. This patient is not simply a "short cervix" patient and not simply a "prior preterm birth" patient. She is an overlap phenotype.

Her recurrence history matters. McManemy taught that risk rises with number, order, and severity of prior preterm births. If there are two prior very early preterm births, the recurrence risk can be much higher than with one remote late-preterm birth followed by a term birth. Her current cervical length adds a present-tense risk signal.

For medical treatment, do not automatically reach for 17-OHPC. Current guidance does not recommend it. Do not automatically substitute vaginal progesterone if the cervix is normal. If the cervix is shortened, vaginal progesterone may be considered, but the full management conversation may also involve mechanical interventions that belong to a different course.

Now compress the course into memory.

Twenty-five millimeters is the short-cervix boundary, but not a biological cliff.

Twenty millimeters or less before twenty-four weeks is the strongest SMFM vaginal-progesterone recommendation zone for the asymptomatic singleton without prior spontaneous preterm birth.

PREGNANT means ten to twenty millimeters, ninety milligrams vaginal gel daily, and preterm birth before thirty-three weeks reduced from sixteen point one to eight point nine percent.

Fonseca means fifteen millimeters or less, two hundred milligrams vaginal progesterone nightly, and spontaneous delivery before thirty-four weeks reduced from thirty-four point four to nineteen point two percent.

Romero two thousand eighteen means the individual-patient-data synthesis: fewer births before thirty-three weeks and mean gestational age at delivery about zero point seven four weeks higher.

Meis means weekly intramuscular 17-OHPC for prior spontaneous preterm birth, with an initially positive result. PROLONG means the confirmatory trial did not reproduce the benefit. Current guidance means 17-OHPC is not recommended, and vaginal progesterone is not its substitute unless a shortened cervix changes the phenotype.

Late initiation means caution. Continuing a regimen that began in the evidence window is not the same as starting after the window. Einum shows a timing gradient. Dominsky and the later asymptomatic cohort abstract do not show time gained after late diagnosis in lower-risk populations.

Finally, remember the endpoint discipline.

Under thirty-seven weeks, under thirty-four weeks, under thirty-three weeks, under twenty-eight weeks, diagnosis-to-delivery interval, mean gestational age at delivery, and neonatal morbidity are not synonyms. A physician who knows the endpoint knows what kind of benefit is being claimed.

The final bedside model is simple, but not simplistic:

Classify the phenotype. Locate the timing. Name the endpoint. Match the evidence. Then prescribe, decline, or counsel with the appropriate degree of certainty.

That is how the details become useful. Not because they were memorized as isolated facts, but because they now form the structure of clinical judgment.